



Exploring the World of Science

Entomology C



Test length: 50 Minutes

Team name: _____

Team number: _____

Student names:

Station 1 _____ / 16

Q1. Insects are an important food source in many cultures. Please list **2 pros and 2 cons** of the practice of using insects as food. (4 pts)

Pros	Cons

Q2. Humans are far from the only creatures for whom insects serve as food! As such, many insects have evolved characteristics or behaviors to keep predators at bay. For each of the following specimens, please **1) identify it** to the lowest taxonomic classification on the 2026 Entomology List AND **2) list one specific defense mechanism** it uses to avoid or deter predators. (12 pts)

Specimen	ID (1 pt each)	Defense Mechanism (1 pt each)
A		
B		
C		
D		
E		
F		

Station 2 _____ / 18

Q1. Identify each of the following insects (Specimens A-J) by the lowest classification on the 2026 Entomology List. (10 pts)

Specimen	Identification	Specimen	Identification
A		F	
B		G	
C		H	
D		I	
E		J	

Q2. Imagine you are a gardener and you are dealing with a severe aphid infestation. Which of the following insects would you introduce into your garden to help **directly** control this problem? Circle the letter(s) of all specimen(s) that apply. (5 pts) ***TIEBREAKER 1***

A B C D E F G H I J

Q3. Some specimens depicted at this station belong to the same order. List the letters of these specimens and identify which order they belong to. (3 pts)

Specimens: _____ Order: _____

Station 3 _____ / 22

Q1. Match the anatomical structure to its correct function: (10 pts)

Anatomical Structure	Function
1. Fat bodies _____	A. Piercing into plant tissue
2. Ocelli _____	B. Respiration
3. Spiracles _____	C. Releasing pheromones
4. Cerci _____	D. Mechanical sensation
5. Mushroom body _____	E. Forms external structures after metamorphosis
6. Pterostigma _____	F. Olfactory processing
7. Halteres _____	G. Simple light detection
8. Imaginal discs _____	H. Metabolism and immune function
9. Stylet _____	I. Stabilizing flight and steering
10. Coremata _____	J. Assisting in gliding

Q2. Insect legs are adapted for a great diversity of different functions. For each specimen below,

1) choose a term from the word bank that best describes the insect's legs, and

2) describe the function that the insect's legs are adapted for (i.e. what the term means). (12 pts)

natatorial	cursorial	saltatorial
foragial	fossorial	raptorial

Specimen	Term (from word bank)	Function
A		
B		
C		
D		
E		
F		

Station 4 _____ / 16

Q1. Please identify each specimen to the lowest classification on the 2026 Entomology List. (3 pts)

Specimen A: _____

Specimen B: _____

Specimen C: _____

Q2. One (or more) of the specimens has a special defense mechanism in which it secretes a foul-smelling substance to deter potential threats. Please **circle** the letter of the specimen(s) this applies to. (3 pts)

A

B

C

Q3. The specimens here are often found in regions with cold winters (like Michigan). Please **circle** the letter of the specimen(s) that generally overwinters in its adult form. (3 pts) ***TIEBREAKER 2***

A

B

C

Q4. Look at the close-up of the hind leg of Specimen C. What is the name of the structure the arrows are pointing to, and how does it help Specimen C achieve its characteristic jump? (2 pts)

Q5. Which continent(s) are each of these specimens native to? (3 pts)

Specimen A: _____

Specimen B: _____

Specimen C: _____

Q6. Please describe the morphology/function mouthparts of each specimen (ex. sponging, piercing-sucking, etc.) (3 pts)

Specimen A: _____

Specimen B: _____

Specimen C: _____

Q7. What does the family of Specimen B get its name from? (1 pt)

Station 5 _____ / 18

Q1. Please identify each specimen to the lowest classification on the 2026 Entomology List. (3 pts)

Specimen D: _____

Specimen E: _____

Specimen F: _____

Q2. Describe the feeding style of each specimen (ex. detritivorous, predaceous, herbivorous, etc) (3 pts)

Specimen D: _____

Specimen E: _____

Specimen F: _____

Q3. What is special about the forelegs of Specimen D and what purpose does this serve? (2 pts)

Q4. Is Specimen E male or female? How can you tell? (2 pts)

Insects of Specimen E's family are popular in Asia among collectors and those who engage in the sport of insect fighting (pitting insects against each other and betting on the winner). There is another kind of insect in the same **order** but different **family** as Specimen E that is valued in very much the same way, as it fights in a *similar* way to how Specimen E does.

Q5. What family might this insect belong to? _____ (1 pt)

TIEBREAKER 3

Q6. Given what you know about the *reason* these insects fight each other, what motivator do you think people introduce to these insects to get them to fight? (1 pt)

Q7. Specimen F often poses a nuisance across the United States when it appears during the fall. What is the main concern people have with this species? (1 pt)

Q8. Specimen F belongs to a species that is highly **polymorphic**. What does this mean? (1 pt)

One famous example of polymorphism occurred in a certain moth population living in Industrial Britain during the 1800s, which underwent changes in the average appearance of individuals due to environmental changes.

Q9. What is the common name of this moth, and what **family** does it belong to? (2 pts)

Common name: _____ **Family:** _____

Q10. What phenotype became more common in this population during the British Industrial Revolution and why? (2 pts)

Station 6 _____ / 22

Q1. Please identify each specimen to the lowest classification on the 2026 Entomology List. (5 pts)

Specimen	Identification
G	
H	
I	
J	
K	

Q2. Please write the letter(s) of the specimen(s) that exhibit true eusociality. (2 pts) _____

Q3. It is very common for other insects to mimic the patterns exhibited by Specimens G, H, and K. What is this type of mimicry called and what is the purpose of this? (2 pts)

Q4. Name two families of insects on the 2026 Entomology List with members that mimic the patterns exhibited by Specimens G, H, and K. (2 pts)

Nursing bees secrete a nutritious substance known as “royal jelly”, which is important in the development of queen bees.

Q5. Nursing bees secrete royal jelly from the: (1 pt)

- A. Salivary glands
- B. Hypopharyngeal glands
- C. Dufour's gland
- D. Nasonov gland

Q6. True or false: all larvae, regardless of caste, are fed royal jelly. _____ (1 pt)

TIEBREAKER 4

All specimens at this station represent orders of insects that are renowned for their excellent flight capabilities.

Q7. Please place Specimens G, I, and J into one of the following categories: (3 pts)

Can Hover	Can Fly Backwards	Cannot Hover OR Fly Backwards	Can Hover AND Fly Backwards

Q8. According to a famous quote, “According to all known laws of aviation, there is no way that a bee should be able to fly. Its wings are too small to get its fat little body off the ground”. And yet the bee flies anyways—by leveraging physics. In 1996, it was discovered that bee flight involves tiny tornado-like airflows that generate lift. These airflows, which are the major mechanism for generating lift in biological flapping flight, are known as: (1 pt)

- A. Wingtip vortices
- B. Leading edge vortices
- C. Horseshoe-shaped vortices
- D. Trailing edge vortices

Q9. What characteristic(s) makes insects in Specimen I’s order particularly strong fliers? Select all that apply. (2 pts)

- A. Ability to control each wing independently
- B. Stabilizing structures on their wings
- C. Excellent 360° vision
- D. Dense innervation of flight muscles with motor neurons
- E. Sensitive proprioceptors at the wing bases

Q10. What family on the 2026 Entomology list is Specimen J often confused with and what are two ways to tell them apart? (3 pts)

Family: _____

1. _____
2. _____

Station 7 _____ / 16

Specimen A pictured at this station is the American cockroach *Periplaneta americana*, which ironically is NOT native to the Americas. Rather, it is native to Africa and the Middle East and was thought to be introduced to the Americas during the transatlantic slave trade.

Q1. What family does this insect belong to? (1 pt) _____

Q2. You may notice that one of the most defining characteristics of cockroaches are the spikes on their legs. What are these spikes called and what function do they serve? (2 pts)

Q3. Cockroaches are often loathed for being dirty and unclean, but they are actually dedicated groomers. What body part do cockroaches groom the most and why? (2 pts)

Q4. Most cockroaches are gregarious, and: (1 pt)

- A. Form large colonies with structured caste systems and task specialization
- B. Live in large groups with no dedicated task specialization
- C. Cooperate in groups of males to find and attract females
- D. Are mostly solitary but congregate occasionally to feed and reproduce
- E. None of the above

Q5. Fill in the blank: Out of the nearly _____ species of cockroaches, about _____ species are considered pests. (2 pts)

Q6. Besides cockroaches' ability to transmit germs, what are the top two health concerns triggered by cockroaches and the substances they produce? (2 pts) ***TIEBREAKER 5***

Q7. Specimen B pictured at this station is in the order Mantodea. Which of the following best describes the systematic relationship between the orders of Specimen A and B? (1 pt)

- A. They belong to the same superorder, Dictyoptera
- B. They belong to the same superorder, Paraneoptera
- C. They belong to separate superorders
- D. They do not belong to any superorder

Q8. In insect systematics, you will often notice that insects are often classified in tribes. Please circle the position that describes where tribes fall in relation to other taxa. (1 pt)

Species • Genus • Subfamily • Family • Suborder • Order

Q9. The units that make up the compound eyes of insects are called: _____ (1 pt)

Q10. Approximately how many of these units make up the eyes of Specimen B? _____ (1 pt)

Q11. What behavior does Specimen B exhibit when threatened? (1 pt)

- A. Attacks the threat with its forelegs
- B. Stays as still as possible to avoid detection
- C. Emits a series of warning clicks
- D. Raises its forelegs and wings to appear larger
- E. None of the above

Q12. True or false: Females of Specimen B cannibalize the male's head during mating in over 50% of cases. _____ (1 pt)

Station 8 _____ / 18

Depicted at this station are the real common names of several moths that belong to the families **Noctuidae**, **Erebidae**, or **Geometridae**.

Q1. Please place each moth into its correct family. (10 pts) ***TIEBREAKER 6***

Noctuidae	Erebidae	Geometridae

Q2. Which of these families were once considered to be a single taxonomic group? (1 pt)

- A. Noctuidae and Erebidae
- B. Noctuidae and Geometridae
- C. Erebidae and Geometridae
- D. Noctuidae, Erebidae, and Geometridae
- E. None of the above

Q3. Continuing from Q2, which of these families split off from the other? (1 pt) _____

Q4. Caterpillars of Geometridae are commonly called “loopers”, while caterpillars of Noctuidae and Erebidae are often called “semi-loopers”. What differentiates loopers from semi-loopers? Select all that apply. (5 pts)

- A. Loopers have more reduced prolegs than semi-loopers
- B. Loopers have four lateral setae while semi-loopers lack them
- C. Loopers have more prominent mouthparts than semi-loopers
- D. Loopers are not hairy like semi-loopers
- E. Loopers move in more of an inchworm-like fashion than semi-loopers

Q5. Many species in Geometridae and some species in Erebidae have a very prominent form of sexual dimorphism. What is it? (1 pt)

Station 9 _____ / 22

Q1. Please identify each specimen to the lowest classification on the 2026 Entomology List. (8 pts)

Specimen	Identification
A	
B	
C	
D	
E	
F	
G	
H	

One adaptation insects have developed for breathing in aquatic environments is by trapping air bubbles or a thin layer of air by their spiracles or underneath their elytra, which allows them to access oxygen when underwater.

Q2. What is this air bubble called? (1 pt) _____

Q3. What anatomical feature is responsible for trapping the air close to the insect's body? (1 pt)

Q4. Insects that use this form of breathing can often stay underwater for long periods of time because they are able to replenish their oxygen supply while underwater. This happens because: (1 pt)

- A. The insects replace their air bubbles with the bubbles trapped on underwater stones, debris, etc.
- B. The air bubble is often replaced by other bubbles generated from turbulence in the water
- C. Oxygen from the water diffuses into the air bubble as the oxygen in the bubble is used up
- D. The insects land on underwater plants and replenish their air bubbles with oxygen produced by the plants
- E. None of the above

Q5. Please circle the letter(s) of the specimen(s) at this station that use this form of breathing. (4 pts)

TIEBREAKER 7

A B C D E F G H

One of the insects at this station is famously known to have two pairs of compound eyes, one that sits above the water's surface and one that sits below.

Q6. Which specimen is it? (1 pt) _____

Q7. These two pairs of eyes are an adaptation to overcome what visual dilemma? (1 pt)

Q8. Fill in the blank: (Females/males) _____ of Specimen B are responsible for carrying eggs on their backs and guarding them. (1 pt)

Q9. Between Specimens D and H, which one is more likely to be attracted to bright lights at night? (1 pt)

Specimen D

Specimen H

Q10. Which specimen represents the family containing the only known oceanic insect? (1 pt) _____

Q11. Members of Specimen C's family release a chemical defense from the prothoracic defense glands, which contain considerable amounts of what type of substance? (1 pt)

- A. Acid
- B. Aldehydes
- C. Enzymes
- D. Steroids
- E. None of the above

Q12. Aquatic insects are important indicator species for water ecosystems. What might it indicate if there are no or very few aquatic insects in a body of water? (1 pt)

Station 10 _____ / 18

Q1. For each disease, please list the letter of the insect that transmits/causes the disease. (8 pts)

- | | |
|----------------------------|-----------------------|
| 1. Tospovirus _____ | 5. Myiasis _____ |
| 2. Chagas disease _____ | 6. Anaplasmosis _____ |
| 3. West Nile virus _____ | 7. Dengue _____ |
| 4. Rice stripe virus _____ | 8. Ehrlichiosis _____ |

Q2. Insects can transmit disease in two ways: mechanical transfer and biological transfer. For the following situations, please identify if this is a case of mechanical or biological disease transfer. (5 pts)

1. A mosquito carrying the protozoan *Plasmodium malariae*, the parasite that causes malaria, bites a human and transmits the parasite to their bloodstream.

2. A cockroach feeds on feces containing *E. coli* and then spreads it to a plate of human food. The human eats the food and gets sick.

3. A mosquito drinks the blood of a rabbit infected with myxomatosis and immediately bites another rabbit, transmitting the virus through the blood on its proboscis.

4. A chrysomelid beetle walks on the leaves of a cucumber plant infected with *Erwinia tracheiphila*, which causes fatal bacterial wilt. The beetle flies off and lands on another cucumber plant, transmitting the bacteria to that plant.

5. A flea feeds on the blood of a rat infected with *Yersinia pestis*, the bacterium that causes plague. After a long time without feeding, the flea bites a housecat. The cat contracts the plague and coughs, releasing infectious droplets in the air that transmit the plague to its human owner.

Q3. What is the most common and debilitating insect-borne disease in the world (and a major focus of medical entomology)? (1 pt)

Q4. Chagas disease is transmitted through: (1 pt)

- A. The vector's bite, which transmits the pathogen to the blood
- B. The vector's digestive juices, which transmits the pathogen to human food when the vector feeds on it
- C. The vector's feces, which transmits the pathogen through physical contact
- D. The vector's defense chemical, which transmits the pathogen when it is sprayed
- E. None of the above

Q5. Why aren't mosquitoes affected by malaria? Select all that apply. (2 pts) ***TIEBREAKER 8***

- A. Their lifespan is shorter than the time required for malaria to take effect
- B. The malaria pathogen has evolved to cause disease specifically in vertebrates
- C. They transmit malaria through the blood on their proboscis, so the malaria pathogen never enters their body
- D. They have immune defenses that fight the malaria infection
- E. Trick question—mosquitoes die from malaria all the time, but there are so many mosquitoes that it's basically unnoticeable

Q6. What's your favorite bug? (1 pt)

Congrats on finishing bug test!!!

